

Underwater Controlled Source Electromagnetic Sensing: Locating and Characterizing Compact Seabed Targets

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Abstract—The detection of conductive or magnetic objects of interest obscured in coastal and marine sediments has motivated the need for advanced marine geophysical technologies suited for relatively small scale search and characterization missions. Targets include both anthropogenic objects such as marine infrastructure associated with undersea cables, seabed foundations for windfarms, and unexploded ordnance and other munitions hazards as well as shallow natural and geologic objects (e.g. freshwater lens, gas hydrates, mineral ore, and heterogeneous sediment deposits). Successes achieved by large scale marine controlled source electromagnetic systems used for hydrocarbon and mineral exploration have paved the way for potential adaptation of sensing strategies and scaled array configurations to detect and characterize these shallower and smaller targets.

Building on established marine electromagnetic theory and based on the use of existing electric and magnetic field sensing designs, we analyze the electromagnetic fields emitted from excited targets in the frequency range between 100 Hz and 200 kHz. We present the results of numerical modeling and experimental studies to develop potential design strategies for implementing both magnetic (B) and electric (E) field sources and receivers. Application of three-dimensional numerical simulations (via the method of auxiliary sources and finite element methods) in addition to one-dimensional analytical models (integral dipole approximations) yield optimal arrangements for a potential advanced electromagnetic sensing system. We study the electromagnetic field distributions

from both electric (voltage-fed dipole) and magnetic field (encased and submerged induction coil) active sources. Controlled source experiments in laboratory and open water settings reveal the effects of signal attenuation, target scattering, and influence of the sea bottom.

Index Terms—unexploded ordnance, magnetometry, electromagnetic induction, underwater positioning

INTRODUCTION AND BACKGROUND

Current underwater geophysical surveys are primarily limited to passive magnetic systems towed from a surface vessel. These systems utilize fluxgate, Overhauser, or atomic magnetometer sensors, often deployed in arrays towed from the stern of small to moderate-size vessels. Active source electromagnetic methods have been utilized to a lesser extent. These systems generally consist of pulsed induction or swept frequency induction coil transmitter and receiver pairs or arrays. A few commercialized systems, mostly based on terrestrial EM systems, have been developed and deployed. While ground-based sensing methodologies have experienced some limited success when applied to the underwater environment, the fundamental electromagnetic characteristics of marine media require an alteration of terrestrial paradigm to fully exploit these conventional methodologies. Specifically, there is much evidence that employing electromagnetic modes that are higher in frequency than those typically used in ground-based sensing will yield greater range and sensitivity for underwater surveys.

We assert that there are significant aspects of the underwater electromagnetic environment not available on land that we should exploit for improved standoff sensing and discrimination. Conventional EMI sensing on land utilizes only magnetic sources and receivers (induction coils). By contrast, underwater/marine geophysics utilizes both electric and magnetic sensors. Some of the earliest work on AC marine electromagnetic methods involved military applications, dating back to early in the last century with work on submarine cables and ship guidance. Fish, such as sharks, surgeons, and catfish sense weak bioelectric fields (~0.1

$\mu\text{V/cm}$) emitted by aquatic organisms - termed “passive electrolocation” - to locate prey. Inspired by this biological system, the US Navy has demonstrated electric field sensors that can detect surface and subsurface craft at distances on the order of hundreds of meters or greater. The use of electric and magnetic fields for vessel detection and characterization continues to be an active area of work in addition to recent applications for submarine and unmanned underwater vehicle communication and underwater radio navigation.

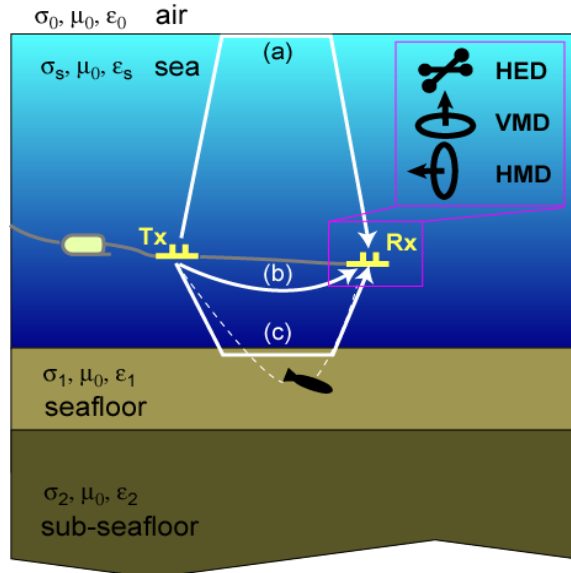


Figure 1. In marine applications, the source is embedded in the conductive medium rather than lying on a conductive half-space. Under these conditions, HED and HMD systems produce and receive both TM and TE modes in the sea (unlike on the surface of the Earth). Total fields represent the superposition of direct, reflected, and interaction with sea surface and seafloor interfaces.

Several marine controlled source electromagnetic methods have been developed (e.g., [2]; [3]) for seafloor sediment mapping (to depths of 100 meters) and larger-scale hydrocarbon reservoir exploration (to depths of 1000's of meters). Systems also exist to measure sub-seafloor resistivity on length scales of a few meters and to depths of $\sim 20\text{m}$ [4].

Four fundamental source types exist: vertical and horizontal magnetic dipoles (VMD and HMD; i.e., loop current source) and vertical and horizontal electric dipoles (VED and HED; i.e., dipole sources). In the surrounding media, each source type generates some combination of transverse magnetic (toroidal mode, TM) or transverse electric (poloidal mode, TE) propagation modes relative to the seafloor.

MODELING AND SIMULATION

Solutions for both electric and magnetic field characterization are well-established in resistive terrestrial media. Numerical and analytical models for sources in conductive (lossy) media have also been developed, but are often validated at much larger scales than those of interest in this study. For example, a number of 1, 2, and 3-D models have been developed for petroleum exploration using

controlled-source EM methods. These models are often only appropriate at large offset ranges or in very deep water. Chave and Cox [1] have developed exact closed-form solutions for vertical and horizontal electrical dipole sources that are appropriate at any range or depth. In addition, Chave [2] has compiled a complete set (all combinations of dipole configurations) of closed-form solutions for both magnetic and electric sources in conductive media. We utilize these formulation for understanding the propagation of magnetic and electric fields near the seafloor. These solutions present models for the various electric and magnetic sources in terms of complex Hankel transforms (integrals of kernel functions multiplied by Bessel functions of the appropriate Green's function expressions). These integrals are solved by approximation or asymptotic solutions and can also be integrated numerically.

Maxwell's equations for an isotropic medium of conductivity σ , permeability μ , and dielectric permittivity ϵ are:

$$\nabla \times \mathbf{H} = (\sigma + i\omega\epsilon)\mathbf{E} + \mathbf{J} \quad (1)$$

and

$$\nabla \times \mathbf{E} = i\omega\mu\mathbf{H}, \quad (2)$$

where the electric $\mathbf{E}$ and magnetic $\mathbf{B}$ fields can be represented by the magnetic vector potential $\mathbf{A}$, with wavenumber k :

$$\mathbf{E} = \frac{i\omega}{k^2} \nabla \nabla \cdot \mathbf{A} - i\omega\mathbf{A} \quad (3)$$

and

$$\mathbf{B} = \mu\mathbf{H} = \nabla \times \mathbf{A}, \quad (4)$$

where $k^2 = i\omega\mu\sigma - \omega^2\mu\epsilon$. The vector potential and associated electric and magnetic fields can be found by solving the Helmholtz equation $\nabla^2 \mathbf{A} - k^2 \mathbf{A} = \mu\mathbf{J}$.

We use both numerical and analytical techniques to simulate the propagation from various source types and scattering of both magnetic and electric field components. Analytical and semi-analytical methods assume isotropic one-dimensional media and represent scattering objects by dipole approximations. These solutions are obtained through the use of complex image theory or asymptotic approximation to the wavenumber domain integral equations.

Our numerical approach utilizes the *Method of Auxiliary Sources* (MAS) to represent both sources and scattering objects in three dimensions [3]. The boundary value problems are solved for EM fields in each domain by finite linear combinations of sources placed along the surfaces of structures or objects. The tangential components of the electric and magnetic fields are continuous across each boundary such that:

$$(\mathbf{E}_2 - \mathbf{E}_1) \times \hat{\mathbf{n}} = 0, \text{ and} \quad (5)$$

$$(\mathbf{H}_2 - \mathbf{H}_1) \times \hat{\mathbf{n}} = 0 \quad (6)$$

In some cases, approximate solutions are advantageous when computation time and complexity precludes a thorough investigation of particular effects. In such situations, we utilize one-dimensional models such as those derived from the theory of complex images (e.g., [4]) and from approximations

of infinite integrals. These formulations are not exact, but provide very efficient and accurate solutions for many scenarios. For example, Bannister [4] extended his complex image theory models for sources and receivers in conductive media for the magnetic and electric dipole configurations of interest here. The formulas are completely general and valid at any frequency range for a flat seafloor double half-space model configuration.

A. Submerged Electric and Magnetic Transmitter Sources

Simulations of active magnetic and electric field sources have been developed and utilized. We first consider electric and magnetic field sources located in a conductive seawater whole space. This simplifies the solution with regard to interaction at interfaces and allows us to simulate the baseline source field for which we will compare to more complex models that incorporate seafloor or sea surface as well as experimental data.

Wait [5] considered the insulated magnetic dipole in a conducting medium, where it was shown that the source fields from the dipole are not dependent on the characteristics of the insulation if the overall diameter of the induction coil antenna is much less than the effective wavelength in the conducting medium. Since the smallest wavelength we consider for nominal seawater (4 S/m) is on the order of 2-5 meters (corresponding to 1 MHz and 100 kHz), it will be assumed that this condition is satisfied. An induction loop transmitter

is represented as a magnetic dipole of strength $i\omega \frac{IdA}{4\pi}$, where I is the current in loop of area dA . It was found that this configuration in seawater will give rise to TE spherical waves both inside (in air) and outside (in the seawater) the insulated cavity where the induction coil resides. Choosing a polar coordinate system and formulating Maxwell's equations in terms of the scalar wave function ψ we have:

$$r^2 \frac{\partial^2 \psi}{\partial^2 r} + \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \psi}{\partial \theta} \right) - k^2 r^2 \psi = 0. \quad (7)$$

Solutions can be formed by applying the boundary conditions, which require the continuity of the tangential electric and magnetic fields across the insulator-sea boundary:

$$H_r = \frac{M(a) \cos \theta}{2\pi r^3} (1 + kr) e^{-kr} \quad (8a)$$

$$H_\theta = \frac{M(a) \sin \theta}{4\pi r^3} (1 + kr + k^2 r^2) e^{-kr}, \text{ and} \quad (8b)$$

$$E_\phi = \frac{-i\omega \mu M(a) \sin \theta}{4\pi r^2} (1 + kr) e^{-kr}, \quad (8c)$$

where $M(a) = IdA \frac{3e^{ak}}{(3 + 3ak + a^2 k^2)}$ is the equivalent magnetic

dipole contained within the insulating cavity of radius a . At any point in space the total time-varying magnetic field will be the vector sum of H_θ and H_r . If we take $\theta = 90^\circ$, i.e. the point of measurement in the plane of the loop, then $H_\theta \rightarrow 1$ and $H_r \rightarrow 0$. The near field attenuation from the dipole reveal sthe

expected $1/r^3$ fall off for the magnetic field and $1/r^2$ fall off for the electric field.

Examples of applying these analytical formulations for a magnetic field source (induction coil) are shown in Figure 2. Here we simulate the distribution of electromagnetic energy in water with conductivity of 4 S/m. A 20 cm radius insulated cylindrical coil is assumed for the magnetic dipole source. The near field distributions for the vertical and radial components of the magnetic field reveal significant attenuation regardless of

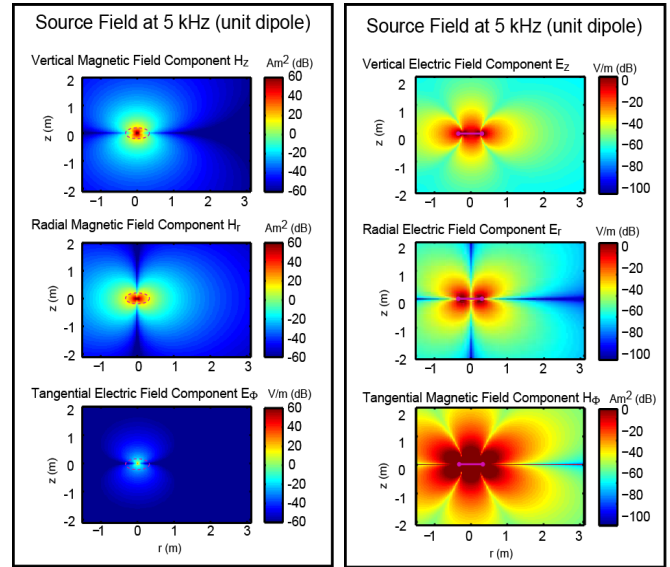


Figure 2. Examples of the application of analytical formulations of a VMD (left) and HED (right) at 5 kHz.

frequency. The tangential electric field component, on the other hand, is highly dependent on frequency, where source fields scale with frequency. At 5 kHz the magnetic field dominates the area around the source. However, at 50 kHz (not shown) the electric and magnetic field components are nearly equivalent in amplitude (in their respective engineering units, Am and V/m).

B. Interaction with the Seafloor

Utilizing the general approach formed for understanding the E- and B-fields due to sources in an infinite conducting medium, we extend our analysis to better understand the interaction of source fields at and across the interface of varying conductivity layers. The transmitting dipole (electric or magnetic) emits electromagnetic signals that diffuse outwards both into the water column and downwards into the seafloor. Interaction with the sea surface and seafloor can both impact the total electric and magnetic fields in the water column where our receivers will be placed.

The full field solutions for a dipole source (electric or magnetic) contained within a conducting medium and surrounded by stratified layers of different conductivity are given by solving Maxwell's diffusion equations in terms of electromagnetic potentials. This theory is very-well established (e.g., [1], [6], [7]). This model accounts for three layers of varying electromagnetic properties: 1) the air layer – represented by a half-space of zero conductivity, 2) the water

layer – a finite layer where conductivity is dominated by the fluid properties, and 3) the seabed – a resistive (less conductive than seawater) unconsolidated sediment layer where objects of interest may reside.

Figure 3 shows an example of the superposition of the components of the electromagnetic field. In this case, we present the radial component of the electric field from a unit moment horizontal electrode source in seawater ($\sigma_1=4$ S/m) over a resistive seafloor ($\sigma_2=0.04$ S/m). The direct propagation imparts a component that dominates in the near field and attenuates exponentially due to the lossy nature of the medium and as $1/r^2$ due to the geometric effect. When $kr < 1$, the direct field is significant and combines with the surface interaction component in an interference pattern, such that the resistive surface interactions (both for resistive seafloor and air interface) tend to absorb EM energy and reduce the total electric field.

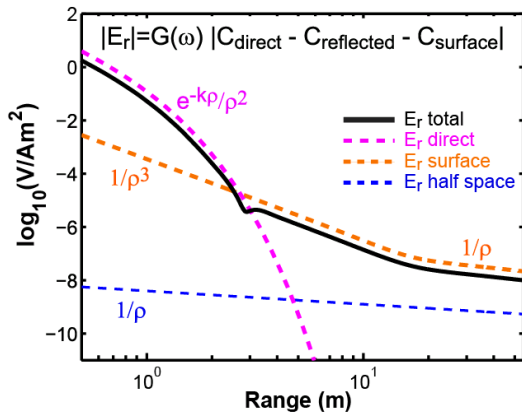


Figure 3. Unit dipole source radial field as determined by the summation of various wave propagation components.

The surface interaction component decreases as $1/r^3$ for small ranges of r and $1/r$ for larger ranges. The “dip” in the total field occurs at $k_2r=1$ where the dominant component transfers from the direct field to the surface interaction fields. The “reflection” component from the existence of the halfspace generally falls off as $1/r$ with range (when approximated using complex image theory). The exact closed form solutions derived by King et al. [8] describe a near range half space response component decreasing as $1/r^{0.5}$ at very short ranges and, then, as $ik/r + (k_1^2/k_2^2r^2)$ at larger ranges.

We consider an example case comprised of a conductive sandwich configuration layer (4-layer model) with a thin highly conductive within a resistive seafloor halfspace. Although the layer interaction is highly dependent on signal frequency, Figure 4 shows the effect of the anomalous conductive layer (the target) on the background layered (air-sea-seafloor) response for 50 kHz broadside and in-line HED configurations. Figure 4 also shows how the anomalous field varies with range from the transmitter. The anomalous field is simply calculated from the ratio of the total field to that of only the background natural media. This illustrates the variability in sensitivity of the E and B field components for both configurations to the inclusion of a buried conductor.

The broadside and in-line configurations generate different

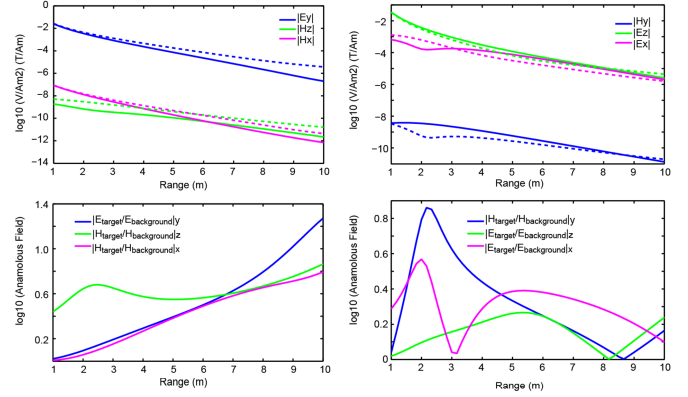


Figure 4. E and B field components and their attenuation with range in the water column over a resistive seafloor with embedded conductivity anomaly. The top plots show the background field components modeled for the 3 layer case absent of the highly conductive layer and the dotted curves show the total response with the conductive layer. Bottom plots show the anomalous fields from the inclusion of the conductive layer. Left – Broadside configuration E and B field components due to a 50 kHz electric field unit dipole transmitter. Right – The field components for the in-line configuration.

anomalous field “move-out” forms. For the broadside configuration the perpendicular (denoted E_y here) electric field component yields the strongest response, but the vertical magnetic field is most sensitive to the conductive anomaly in the near range (<6 m). For the in-line configuration, the electric field components generate the strongest response with weaker overall responses from the orthogonal magnetic component.

C. Scattering Responses from Compact Metallic Targets

The detection and localization of submerged compact objects in conductive water relies on our ability to sense scattered fields induced by either magnetic or electric primary fields. The scattered field is determined first by simulating the incident field at the location of the scattering object, and then generating the surface currents and representing them as magnetic and/or electric dipole moments. Reflections between the scattering object and the surface of the conducting medium must be accounted for. We consider cases when the object is in an infinite conducting medium and when it is buried beneath or lying on a resistive seafloor.

For the case of a sphere excited by a VMD source in an infinite medium we start from the classical analytical formulation of Wait [9]. This solution is extended to account for the finite conductivity of the medium and interaction of E and B field components. We utilize the solution in terms of the scattered vector potential:

$$\mathbf{A}^s = \frac{\mu}{4\pi r^3} \left(\frac{1+k_2r}{1+k_2a} \right) e^{-k_2(r-a)} (\beta \mathbf{H} \times \mathbf{r}), \quad (9)$$

where the frequency response of a sphere can be represented by

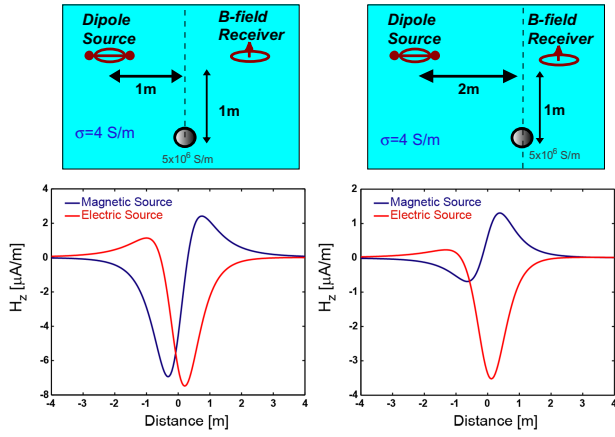


Figure 5. The vertical component of the magnetic field scattered from a 5 cm radius steel sphere. (Left) The center of the source electrode (red) and induction coil (blue) are both centered at -1 m from the target. (Right) The amplitude of the scattered magnetic field generated from the electric source is nearly double that generated by the magnetic source at 2m offset.

$$\beta(\omega, a) = 2\pi a^3 \frac{3\mu(k_1 a \coth(k_1 a) - \mu(k_1 a)^2)}{(k_2 a)^2 [k_1 a \coth(k_1 a) - 1] + \mu(k_1 a)^2}, \quad (10)$$

where a is the radius of the sphere. Similar response functions can be formulated for flat plates and solid cylinders. The magnetic and electric fields can then be obtained from the vector differential equations:

$$\mathbf{H} = \frac{1}{\mu} \nabla \times \mathbf{A} \quad (11)$$

$$\mathbf{E} = \frac{i\omega}{k^2} \nabla \nabla \cdot \mathbf{A} - i\omega \mathbf{A}. \quad (12)$$

We investigate the secondary magnetic fields from a conducting and permeable sphere in a conducting saltwater host medium. Figures 5 and 6 show profiles across the plane of the transmitter for the vertical and lateral magnetic field components scattered from a 10 cm diameter sphere with conductivity of 4×10^6 S/m and magnetic permeability of 100. The B field source is comprised of a 1m X 1m induction loop transmitter and the electric field dipole is represented by two 1 m segments of bare wire.

Comparisons of the scattered magnetic fields generated from an E field source relative to a B field source are shown. Equivalent power from a 1 A input current is used to simulate the incident source field, which in turn, generates the scattered fields. At a source-target offset of 1 m, the two sources lead to scattered magnetic fields of roughly the same strength. However, as the transmitter moves further away from the target (i.e., increasing range), the E field source tends to generate stronger scattered magnetic fields. We mainly attribute this to the increase in the source field strength incident on the target from the E field source relative to the B field source.

EXPERIMENTAL DATA COLLECTION

To provide an experimental basis for comparison and validation of modeling results, we conducted both laboratory

and open water data collections. Initial measurements were taken in a controlled saltwater test tank, primarily to evaluate low-noise electric field sensor response to nearby metallic targets. Subsequent testing in two different open water environments provided freedom from the constraints of testing in a water tank and to reduce the effects of noise associated with the laboratory.

A. Laboratory Experiments

Laboratory experiments were conducted in a 3.1m by 1.2m by 1.2m non-conducting test tank filled with saltwater with conductivity of 5.2 S/m. We acquired data with a set of 3-axis Ag/AgCl electrode receivers. Initial receiver tests utilized a very weak current source via a set of silver electrodes. Although the source was limited to very low power levels (millivolt inputs), direct transmission of the electric field was observable up to 3m from the source. Noise characterization of these and similar sensors were limited by their respective low-noise amplifiers to 1-2 nV/ $\sqrt{\text{Hz}}$ /m over the band of 100 Hz to 20 kHz.

In preparation for open water tests, we also conducted a series of laboratory experiments to establish the electromagnetic characteristics associated with different electrode transmitter source configurations. One of the most important relationships established through these experiments was the effect of the electrode geometry (e.g., length, diameter, spacing) on the overall circuit resistance. Because the electrodes are galvanically coupled to the water, the impedance of the water column contributes an effective resistance to the current path. Thus, the contact resistance and saltwater impedance tend to influence the amount of current that can be driven through the electrodes.

Through controlled experiments, we were able to determine some of the requirements for minimizing the electrode circuit impedance (thus maximizing source current) through optimization of source configurations. Scaled versions of fieldable electrodes ranging in size from 2.5 to 5 cm in diameter were submerged at fixed depths in the water tank. Using a highly sensitive inductance-capacitance-resistance (LCR) meter, we measured the effective resistance between electrode pairs. A significant drop in resistance (0.6 to 0.2 Ohms) was observed between the 2.5 cm and 5 cm diameter electrodes. We did not observe a significant reduction in contact resistance using 5cm diameter hollow cylindrical electrodes relative to 5cm diameter solid graphite electrodes. We did, however, observe relatively systematic reduction in resistance as distance between the electrodes were decreased. As a final test, we evaluated the possibility of adding electrode pairs in parallel as a means for reducing the circuit impedance. Observations from these tests indicated that circuit resistance was reduced by more than 40% when 4 electrodes were utilized in parallel relative to a single pair.

B. Open Water Experiments

Two sets of open water experiments were conducted to develop a fundamental understanding of magnetic and electric field responses to varying marine environments, to metal targets of interest, and to proximity to the seafloor and air-sea

interface. Furthermore, analysis of data from these experiments provided validation of analytical and numerical modeling and observations of small-scale (1-10 meter) range attenuation for electric and magnetic field receivers.

1) *Controlled Brackish Water Tests: Choctawhatchee Bay, FL:* Initial open water experiments were conducted in shallow water (up to 4m water depth) near a dock facility off Okaloosa Island, Florida in the Choctawhatchee Bay. Data acquisition and power were supplied topside and utilized the dry environment of the dock. Pressure, temperature, and conductivity were measured with a handheld CTD sonde (YSI Castaway). The conductivity in this area, 2.6 ± 0.4 S/m, is lower than that of ocean water due to freshwater inputs upstream in the bay. Water temperatures between 8 and 9 degrees C generated a need for wetsuits for personnel working in the water during experiments.

For this data collection we applied both electric and magnetic field sources. The magnetic field source consisted of a 1m x 0.5m custom wound induction coil comprised of a 10 turns of 12 AWG single strand, insulated, copper wire. We utilized this source primarily with the coil flat such that the magnetic dipole was oriented vertically in the water column (i.e., a vertical magnetic dipole or VMD). Alternatively, in a few instances the magnetic dipole was oriented horizontally (HMD). The electric field source consisted of a pair of 8cm diameter by 40 cm long graphite electrodes, which were generally oriented in-line and parallel to the seafloor (i.e., a horizontal electric dipole or HED). Photographs of the induction coil and electrode transmitters used are shown Figure 6.

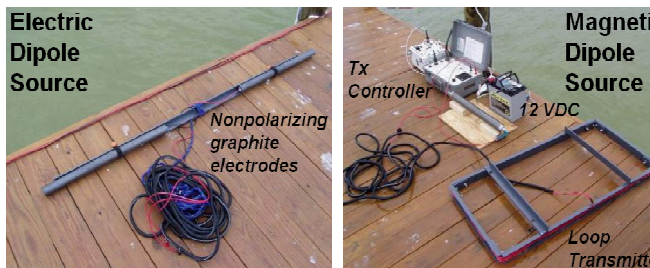


Figure 6. Photographs of the custom electric and magnetic dipole transmitter sources used during experiments.

The electric dipole source reflects E-field signals that are proportional to the gradient of the voltage potential across the electrodes. The magnetic dipole source generates E-field signals that are proportional to the time rate of change of the vector potential rotational magnetic field. The primary E-field signal source for the horizontal electric dipole are galvanically driven currents, whereas the primary E-field signal from the vertical magnetic dipole is driven by inductively coupled currents.

For an electric field receiver we used the Ultra Electronics PMES Ag/AgCl electrodes and integrated ultra low-noise pre-amplifiers. The magnetic field receiver used was the KMS Laboratory for Electromagnetic Innovation (LEMI) LEMI-134 wideband B-field sensor. This sensor is capable of resolving magnetic fields of a few hundred picoTesla over the frequency

range of 17Hz to 280 kHz. We also acquired baseline data with two commercial terrestrial EM systems (Geophex GEM-3 and Geonics EM-61 MkII) by encasing them in a sealed pressure vessel.

Our custom HED and VMD transmitters were submerged in the water column at various depths from the water surface to the seafloor. The transmitters were held stationary in place while the B-field receiver measuring the total vertical magnetic field was systematically moved from 2 m out to 7.5 m away from the center of each transmitter type. Signal attenuation was sufficient to record raw voltage output of over 1 V out to 5 m for both source types.

These surprisingly strong magnetic fields matched our model results (Figure 7). Both the model and the measurements indicate a crossover in the field intensity curves at a distance of approximately 4.5 m from the sources. After this point, the rotational magnetic field generated by the electric dipole is stronger than the magnetic field produced by the magnetic dipole source (given both sources are carrying equal current).

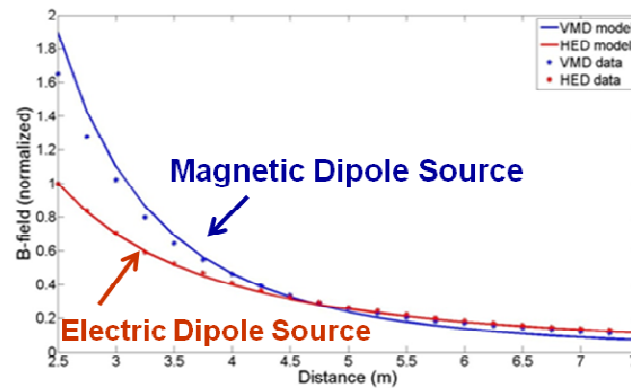


Figure 7. Calculated rotational and vertical magnetic field intensities (solid lines) compared to measured rotational and vertical magnetic field intensities (dots). Measurements confirm model predictions that indicate slower decay of rotational magnetic fields.

In addition to primary field range characterization, we also evaluated signal-to-noise ratio (SNR) of scattered fields from compact metallic targets induced by the two types of sources (HED and HMD). For each source, we measured SNR for scattered fields generated by targets that were maximally coupled with the source. We isolated the scattered field generated by the target by subtracting the background response (i.e., no target present) from the total response (i.e., target present). We measured the scattered field magnitude for the target located at several offsets from the B-field receiver. For close ranges (50 cm), the HMD configuration provided a higher SNR (52 dB for the HMD compared to 48 dB for the HED); however, at greater target offsets (>1 m), the HED configuration provided higher SNR (45 dB for the HED compared to 43 dB for the HMD). Figure 8 shows the measured scattered field response for each source with the target located at a 50 cm offset and at a 1 m offset.

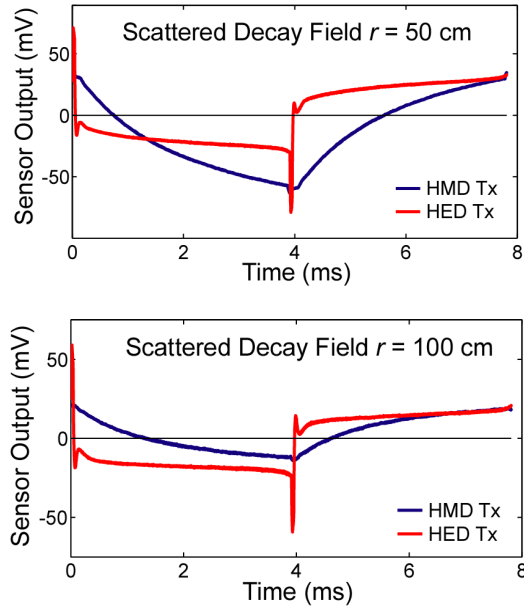


Figure 8. Scattered field measured in sensor voltage for the two configurations. At a 50 cm standoff (top) from the receiver, the target produces a stronger response when excited by the HMD source; however, at a 1 m standoff (bottom) the HED source elicits a stronger scattered field.

2) *Towed System Seawater Tests: Woods Hole, MA:* The initial proof-of-principle experiments elucidated several design requirements for an operational prototype. Most notably, the application of an E field (electric dipole) source offers significant range capabilities when compared with those of the B field (magnetic dipole) source. In simple terms, the E field source acts as an apparent finite or infinite line current source depending on range versus the electrode separation distance. Conversely, the B field source acts as a current loop source. The magnetic fields generated by each source type are determined using the vector potential equation for a lossy medium:

$$\mathbf{A} = \frac{\mu I}{4\pi} \int \frac{e^{-jkR}}{R} d\mathbf{l}' \quad (13)$$

where I is the source current. The wavenumber, k , accounts for the effects of both displacement and conduction currents in the medium surrounding the source.

Equation 13 yields the complete solution for both line (E field) and loop (B field) sources immersed in sea water. From both our modeling studies and our initial experiments, it is evident that the line source offers several benefits for marine sensing applications. Unlike terrestrial resistivity applications where the electrode/soil interface provides a significant contact resistance, galvanic coupling of the line source electrodes to the sea water (at this site ~ 3.9 S/m) enables injection of large currents at relatively low source voltages. This coupling facilitates line source configurations that can be easily deployed in the marine environment. Large electrode separations can be achieved with little hydrodynamic resistance if a streaming configuration is applied. Once galvanic coupling of the electrode to the water column is achieved, high currents

can be driven over large separation distances to produce line sources with very strong moments.

We conducted a subsequent set of experiments to prove some of the operational concepts associated with an E field source configuration. Our conceptual design comprised an array of four parallel 2.5m line sources. Each line source was terminated with a 5 cm diameter graphite electrode to couple the current source to the surrounding water column. To measure the decay of secondary fields generated by metallic targets, we used an array of three-axis induction coil receiver cubes. A receiver was placed in between each line source and oriented such that the horizontal axes of sensitivity were rotated 45 degrees with respect to the line source axis (Figure 9).

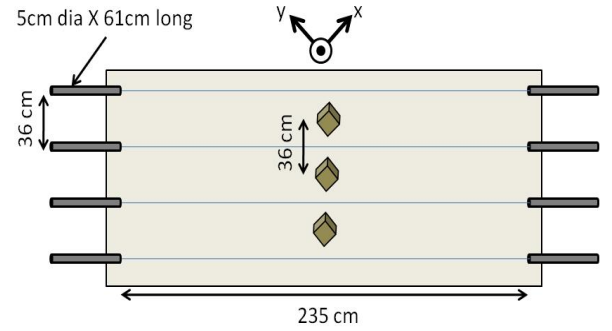


Figure 9. The E field transmitter design comprised 4 parallel line sources terminated with graphite electrodes. A three-axis induction coil receiver cube was placed between each line source to measure the magnetic field decays generated by nearby conductive/magnetic targets.

We implemented this source/receiver design on an experimental platform suspended beneath two pontoons. The platform was fastened to the flotation system using non-conducting PVC tubes that could be raised or lowered to facilitate testing at different water depths (Figure 10).



Figure 10. Experimental platform suspended from a non-metallic flotation system. The electrode line sources and receiver cubes were fastened to the submerged platform. The system was towed from a small research vessel.

The line sources were energized by a high current transmitter driver. We applied a pulsed waveform scheme where the induction coil receivers were synchronized to begin recording magnetic field decay signals 100 microseconds after the transmitter line current was shut off. The measurement period extended to approximately 5 milliseconds after the source current termination. A differential RTK GPS was mounted on the flotation system to record the location of the

sensor platform. Figure 11 shows the complete experimental system towed behind the research vessel.

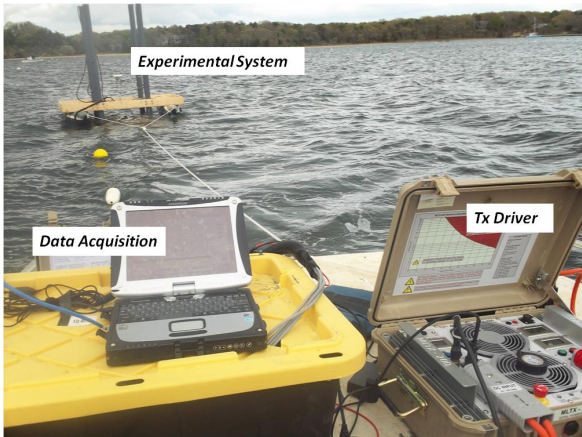


Figure 11. Experimental setup. Data acquisition and transmitter driver electronics were kept topside. Transmitter source current, analog receiver signals, and GPS output were transferred between the vessel and the experimental system via a power/communications tether.

We collected several data sets over a variety of submerged metallic targets. The sensor platform was towed at multiple sea floor standoffs to determine range of detection. Targets were placed either on the sea floor surface or buried such that the top portion was flush with the sea floor. Targets included a 105mm projectile, a 60mm mortar, a 3" diameter steel tube, a 2" diameter Aluminum tube, and a 3" diameter steel sphere (Figure 12).



Figure 12. Targets ranged in size from a 3" steel sphere to a 105mm projectile.

Data sets were acquired in survey fashion with the sensor platform towed along transects above the targets. Using the GPS data, parallel transects were spatially registered and merged to generate coordinates for each receiver measurement recorded during the survey. The receivers produced a transient voltage signal corresponding to ambient magnetic field decays. Background decays were recorded with the receivers located away from targets to provide a reference signal (Figure 13).

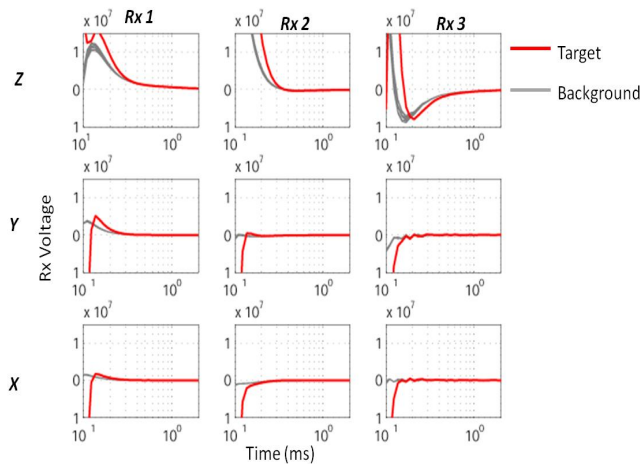


Figure 13. Receiver channel decays. Red lines correspond to measurements recorded above a target (60mm mortar); grey lines correspond to background measurements.

Each receiver provided three channels; one for each orthogonal sensing axis. During post-processing, we integrated a discrete portion of each receiver channel decay to produce a magnetic field amplitude value. These values were subsequently registered with the positioning data to produce magnetic field anomaly maps (Figure 14).

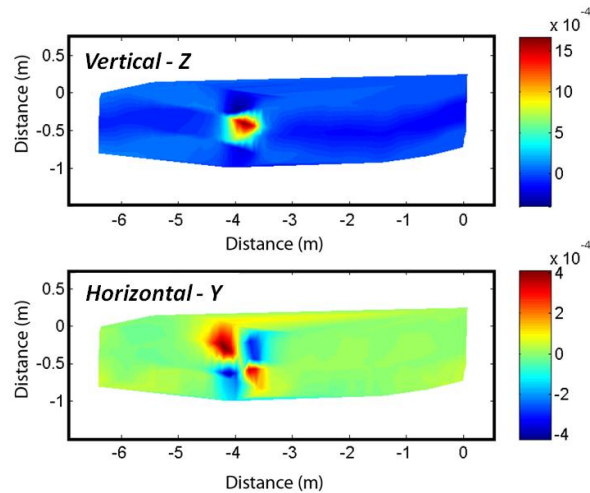


Figure 14. Magnetic field maps generated from measurement transects recorded above a 3" steel tube located on the sea floor. The sensor array was approximately 35 cm above the sea floor during this survey. The top plot shows the amplitude recorded by the vertical measurement axis and the bottom plot shows the amplitude recorded by one of the horizontal measurement axes.

We assessed the detection capabilities of the E field line source modality through a number of experiments. We identified methods that exploited marine electromagnetic phenomenology to boost range performance. For example, increasing the separation between the electrodes can significantly increase the transmitter field at large distances from the line source. Because the water column offers very low resistance to the electrode return currents, we were able to maintain very high current levels at large electrode offsets. Specifically, we measured only an approximate 10 percent decrease in source

current when the electrodes were moved from a 2m separation to a 10m separation. These operational experiments provided insight into the range capabilities of the E field source modality and indicated the potential to exceed the performance of conventional B field loop sources.

SUMMARY AND CONCLUSIONS

A combination of modeling and experimental data collection and analysis were used to gain new insight into active source electromagnetics for underwater characterization. Through simulation of various model scenarios, we showed that the near field distribution of electromagnetic energy around magnetic and electric dipole sources is highly dependent on frequency and geometry. At low frequencies, the VMD generates strong magnetic fields and a weaker electric field component. As frequencies increase the relative strength of the electric field increased significantly. At ~50 kHz the electric field imparts comparable intensities and should be accounted for in models of target or seafloor interaction. Similarly for scattered secondary fields, we show that a VMD source excites mainly magnetic field scattering at frequencies <10 kHz. At frequencies higher than 10 kHz, however, the relative strength of the electric field components cannot be ignored.

Simulations of electric and magnetic field interaction with the sea surface interface and seafloor sediments of varying conductivity show that energy is absorbed by these resistivity boundaries. The total electromagnetic field in the conducting medium where the sources and receivers reside is a superposition of the effects of direct propagation, interaction with the sea surface and seafloor interfaces, and reflections from the interfaces. Comparisons of the various E and B field components from VMD and HED transmitters reveal varying degrees of interaction with the boundaries.

Experimental data collection was conducted with both E and B field transmitters and magnetic field receivers. The electric dipole source generates E-field signals that are proportional to the gradient of the voltage potential across the electrodes, and the magnetic dipole source generates E-field signals that are proportional to the time rate of change of the vector potential rotational magnetic field. The primary E-field signal sources for the horizontal electric dipole are galvanically driven currents, whereas the primary E-field signal from the vertical magnetic dipole is driven by inductively coupled currents. From these experiments we observed relatively strong magnetic fields at transmitter-receiver separation distances >2 m.

Because galvanic coupling from an electric dipole source was found to generate strong and widely distributed primary field conduction currents that fall off as $1/r^2$, we think that a carefully designed E-field transmitter will generate adequate excitation of targets at standoff distances up to 3 meters. Coupled with potential for improved form factor and reduced hydrodynamic drag, we contend that this source may ultimately outperform an induction coil transmitter as a source when cost effectiveness, ease of deployment, and source strength at range are considered.

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